

Internet Appendix for “The Market’s Off-Frontier Component: Predictable but Not Priced”

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This Internet Appendix reports supplementary tables and details. Section and table references of the form 2.5 or Table 5 point to the paper. All variables, samples, and estimators are as defined there.

IA.A The off-frontier component across factors and frontier spans

Table IA.1 reports the across-factor check of Section 4.2 of the paper. Forming $\eta_i = f_i - w_i R^{e*}$ for each of the five factors, only the market’s (and, more weakly, SMB’s) off-frontier component predicts the smallest size decile with a positive sign.

Table IA.2 reports the frontier-span check of Section 5.3 of the paper. Whether the frontier is FF3, FF5, FF6, or FF5 plus the 49 industry portfolios, η remains essentially the same object, still forecasts the smallest size decile out of sample, and stays largely unpriced against the size–value kernel.

Out of sample means two different things, and the recursion of Section 2.4 of the paper delivers only one of them. Every predictive coefficient, frontier weight, and split uses data through t alone, so the results are fully out of sample in the coefficient sense. The factor *definitions* are not. The breakpoints are reconstituted point in time each June, but the choice of which characteristics form the factors was made with knowledge of the history, in 1993 for the three-factor model and 2015 for the five. Two checks bound what this remaining look-ahead can contribute. First, the location result survives when construction and evaluation

Table IA.1: Off-frontier component of each factor as a predictor of the size cross-section

factor f_i	$\text{corr}(\eta_i, \text{Mkt})$	decile-1 OOS R^2	Clark–West	in-sample t (dec. 1)
Mkt–RF	0.88	3.65%	3.9	5.53
SMB	0.17	1.35%	2.2	4.25
HML	−0.41	−0.13%	1.1	−3.24
RMW	−0.43	1.33%	2.2	−4.07
CMA	−0.61	0.29%	1.6	−4.32

Notes. For each factor, $\eta_i = f_i - w_i R^{e*}$ is its off-frontier component (orthogonal to the same FF5 frontier, $E[\eta_i] = 0$). Each row predicts the smallest size decile’s month- $t+1$ excess return on $\eta_{i,t}$: OOS R^2 recursive versus the prevailing mean, in-sample t Newey–West (12 lags). The correlation $\text{corr}(\eta_i, \text{Mkt})$ is contemporaneous. Only the market’s (and, more weakly, SMB’s) off-frontier component predicts with a positive sign; HML, RMW, and CMA enter negatively, tracking their negative market correlation. Sample 1963:07–2026:04.

Table IA.2: The off-frontier result across frontier spans

frontier span	$\text{corr}(\eta, \eta_{\text{FF5}})$	$\text{corr}(\eta, \text{Mkt})$	decile-1 OOS R^2	CW	priced share
FF3 (Mkt, SMB, HML)	0.80	0.70	3.33%	3.8	0.0%
FF5 (baseline)	1.00	0.88	3.65%	3.9	2.6%
FF6 (FF5 + Mom)	0.96	0.92	3.65%	3.9	4.7%
FF5 + 49 industries	0.91	0.96	3.03%	3.4	9.5%

Notes. For each span, $\eta = \text{Mkt} - \text{RF} - w R^{e*}$ is rebuilt with R^{e*} the projection of **1** onto that span. The decile-1 OOS R^2 and Clark–West are recursive forecasts of the smallest size decile (as in Table 1 of the paper); the priced share is $\text{corr}(\eta, \text{MVE})^2$ against the same 25 size–book-to-market SDF, the size–value kernel of Table 5 of the paper, not the R^{e*} frontier whose 0.0% entry there is mechanical. Sample 1963:07–2026:04.

both postdate the definitions. Rebuilding η from the FF3 frontier and evaluating only on 1994:01–2026:04, months entirely after those definitions were fixed, gives a smallest-decile out-of-sample R^2 of 2.7% (Clark–West 2.9 over 388 months), and on that window alone the residual remains unpriced, with a priced share below 1% and $\rho \leq -0.09$ against both the size–value and disjoint kernels. Second, the forecast claim requires no factor construction at all. From the standpoint of a forecaster η is the lagged market (Section 4.2 of the paper), and the lagged market, a predictor with no constructed inputs, earns 1.8% (Clark–West 2.4) on the same window. The five-factor counterpart window, evaluation after the 2015 definitions, shows nothing (Clark–West 0.5 over 130 months), but so does the definition-free lagged market there (0.2), which places that null with the post-decimalization attenuation of Section 8.2 of the paper, not with the construction.

IA.B Conditional timing under Ferson–Siegel optimal weights

The demeaned overlay of Section 6.1 of the paper is transparent but not the unconditionally efficient use of the signal. As a robustness check I form the Ferson and Siegel (2001) optimal conditional weights, $w(z_t) \propto \Sigma^{-1}\mu(z_t)/(1 + \mu(z_t)^\top \Sigma^{-1}\mu(z_t))$, with $\mu(z_t)$ the recursively estimated conditional mean of the ten deciles and Σ a constant (κ -shrunk) covariance. The Ferson–Siegel portfolio earns an out-of-sample Sharpe ratio of 0.70, above both the naive linear rule $w \propto \Sigma^{-1}\mu(z_t)$ (0.62, since the optimal rule’s attenuation in extreme-signal states helps even out of sample) and the full- η demeaned overlay (0.59). The location is unchanged by the richer rule. Conditioning the optimal weights on the unpriced component η_u alone delivers 0.67, on the priced component η_p alone 0.30. A paired stationary-block bootstrap of the Sharpe difference (as in the overlay tests) puts the gap at 0.37 with a 95% interval of $[0.03, 0.71]$ ($p = 0.014$), so it is not noise. The priced-direction rule does retain some absolute value under the richer weights (its 0.30 exceeds the 0.16 implementable static tangency), so optimal weighting extracts something from the priced state. The located gap between the directions nonetheless survives. The conditional-efficiency gain is *larger* under optimal weights and the bulk of it still lives in the unpriced span. The full unconditional tangency, by contrast, falls to an out-of-sample Sharpe of 0.16 once Σ^{-1} is estimated on the near-collinear deciles. Its Sharpe rests on the small, imprecisely estimated cross-sectional spread in mean decile returns, exactly what collinearity makes fragile, while the conditioning gain draws on the strong time-series signal ($t = 5.5$), which even the Ferson–Siegel rule exploits without leaning on that spread. Collinearity therefore sinks the static tangency but not the conditioning gain, which is why the transparent overlay, not the estimated tangency, remains the primary object in the paper.

IA.C Mechanism details: sentiment, industries, and firm size

This section collects the exhibits behind the mechanism tests of Section 7.1 of the paper. Table IA.3 reports the sentiment interaction, Table IA.4 the industry η -loadings by firm-size tercile, and Table IA.5 the ranking of the 49 industries by average firm size.

Firm size is Kenneth French’s *Average Firm Size* for each of the 49 industries, the month-end total market equity of the industry divided by its number of firms, in \$M. I summarize each industry by the geometric mean of its monthly series. This tempers the influence of

Table IA.3: Sentiment and the η lead-lag

<i>Panel A: interaction regression, $R_{\text{dec1},t+1}$</i>			<i>Panel B: η slope by regime</i>		
	coefficient	t		slope	t
η	0.0145	5.34			
sentiment	−0.0071	−2.73	low sentiment	0.0195	4.45
$\eta \times$ sentiment	−0.0048	−2.29	high sentiment	0.0093	3.96

Notes. Macro-orthogonalized Baker–Wurgler index, 1965:07–2023:12 (702 months), 52 months short of the main 1963:07–2026:04 sample, 24 at the start and 28 at the end, where the sentiment index is unavailable. Newey–West (12-lag) t -statistics. Both η and the sentiment level are standardized to unit variance, so slopes are per one standard deviation of the regressor. Regimes split at the median sentiment level.

Table IA.4: Industry η -loadings and firm size

firm-size tercile	mean predictive η -loading	mean $ t $
small-firm	0.0073	2.38
mid	0.0046	2.00
large-firm	0.0022	1.18
corr(loading, log firm size)	−0.61	

Notes. For each of the 49 industries, the predictive loading is the slope from regressing the industry’s month- $t+1$ excess return on standardized η_t (Newey–West 12-lag t -statistics); firm size is the time-series average of the industry’s log average firm size. The tercile-portfolio regressions quoted in Section 4.1 of the paper ($t = 2.8$ to 0.9) are a different statistic from the mean of industry-level $|t|$ shown here. Sample 1963:07–2026:04.

unusually high *months* in the time series, but it does not remove the influence of large constituent firms *within* a month. To test whether industry A ’s firms are smaller than B ’s, I form the monthly log gap $\log(\text{size}_{B,t}) - \log(\text{size}_{A,t})$ and report the Newey–West t -statistic (12 lags) on its mean. The pairing removes the common market component, and the HAC correction handles the series’ persistence. French’s measure is an *arithmetic per-firm mean* (total industry market equity divided by the number of firms, not a value-weighted mean or a median), so within a month its dollar level overweights the largest constituent firms. The geometric time-series averaging does not undo that, but the cross-industry *ranking* is not sensitive to it.

Table IA.5: The 49 industries ranked by average firm size

#	ind.	\$M	#	ind.	\$M	#	ind.	\$M
1	RIEst	184	18	Ships	610	35	Paper	1,378
2	FabPr	192	19	Meals	627	36	Insur	1,383
3	Rubbr	237	20	Fun	635	37	Food	1,395
4	Txtls	280	21	ElcEq	643	38	Chems	1,487
5	PerSv	296	22	Steel	646	39	Autos	1,598
6	Toys	325	23	MedEq	651	40	Util	1,809
7	Cnstr	334	24	Mach	693	41	Drugs	1,829
8	Whlsl	343	25	Books	701	42	Hshld	1,863
9	BusSv	395	26	Other	742	43	Soda	1,995
10	Clths	430	27	Fin	757	44	Aero	2,063
11	Agric	435	28	Mines	799	45	Hardw	2,220
12	Hlth	447	29	Banks	849	46	Oil	2,283
13	Coal	491	30	Chips	906	47	Beer	3,015
14	BldMt	510	31	Trans	929	48	Telcm	3,649
15	LabEq	525	32	Boxes	1,086	49	Smoke	6,470
16	Softw	544	33	Rtail	1,304			
17	Gold	567	34	Guns	1,336			

Notes. Average per-firm market equity (\$M), geometric mean of the monthly series over 1963:07–2026:04, the same sample as the mechanism tests and the -0.61 correlation. Labels are Kenneth French’s 49-industry abbreviations. Smaller-firm industries (low #) carry the larger predictive loading on η (cross-industry correlation with log size -0.61 ; Table IA.4).

IA.D A second-moment footprint: forecasting cross-sectional dispersion

The state variable of this section is the volatility of η , $dis = sd(\eta)$. Because the components of (1) in the paper are orthogonal, $var(Mkt - RF) = w^2 var(R^{e*}) + var(\eta)$, so dis is the part of the market’s volatility off the frontier, and that part is most of it. The off-frontier residual carries 77.2% of the market’s return variance, the frontier-spanning part the remaining 22.8%. This share and the 0.88 market correlation of Section 2 of the paper are the same fact, not two. Because η is a sub-component of the market, $corr(\eta, Mkt) = sd(\eta)/sd(Mkt) = \sqrt{0.772} = 0.88$. Because dis serves as a monthly state variable, I measure it at the monthly frequency as a realized quantity from daily data (formulas below).

The off-frontier component forecasts not only the direction of the cross-section’s adjustment (Section 4.2 of the paper) but its dispersion, a secondary corroboration of the same gradual-adjustment mechanism (I use “diffusion” loosely below for that differential-speed adjustment, informational or liquidity-paced, which Section 7.1 of the paper does not separate). Adjustment at different speeds across assets shows up as cross-sectional dispersion in the following period. Let CSV_t be the cross-sectional variance of the 49 industry returns in month t (forward, realized, and distinct from the backward-looking covariance level) and dis_t the realized monthly volatility of η . Because dis_{t-1} uses month- $(t-1)$ daily returns and CSV_t uses month- t returns, predictor and target share no daily observations.

In sample, dis predicts CSV beyond lagged dispersion, market realized variance, and aggregate idiosyncratic volatility (AIV, the average within-month daily market-model residual variance across the 49 industries). With lagged dispersion and AIV as controls, dis enters at 0.27 ($t = 4.7$) and AIV at 0.47 ($t = 8.4$), with lagged dispersion insignificant (Table IA.6, Panel A). Adding market realized variance as a fourth regressor leaves dis significant (0.47, $t = 3.9$). Out of sample, adding dis to the autoregressive benchmark raises the dispersion-forecast R^2 from 30% (AR alone, not tabulated) to 40%, and to 46% alongside AIV. The Clark–West statistic for dis is 7.1 over the autoregressive benchmark and 4.1 over the benchmark that already includes AIV (Panel B). The improvement survives dropping the 2008–09 and 2020 windows (6.9 and 3.9) and dropping all NBER recession months (6.6 and 4.1). An in-sample recession interaction shows dis ’s slope is, if anything, weaker in recessions.

The increment is modest and should be read with care. Adding dis to the AR+AIV benchmark raises the out-of-sample R^2 from 43.3% to 45.7% in the full sample (about two points), and dis and AIV are 0.70 correlated, so the split between them is fragile. In the recession subsample ($n = 40$) dis adds only 0.3 points and the dis +AIV Clark–West statistic falls to 0.6. I read it as a small, robust, non-crisis-driven increment consistent with diffusion

(η tracing the first and second moments of one gradual adjustment; Stivers, 2003; Connolly and Stivers, 2003; Garcia et al., 2014), not as an independent result.

Realized measures. Within month t : $RV_t = \sum_d (\text{Mkt} - \text{RF})_d^2$; $dis_t = (\sum_d (R_d^e c_t)^2)^{1/2}$ with c_t the linear weight vector of Appendix A of the paper; $AIV_t = \frac{1}{N_t} \sum_i (\frac{1}{d_t} \sum_d \hat{e}_{i,d}^2)$, where $\hat{e}_{i,d}$ is the daily residual from regressing industry i 's excess return on $\text{Mkt} - \text{RF}$ within the month, d_t is the number of trading days, and the outer average runs over the $N_t \leq 49$ industries with complete daily data that month; and $CSV_t = \text{var}_i(R_{i,t})$ across the 49 industries. The dispersion regressions are in logs.

Table IA.6: Forecasting cross-sectional dispersion

<i>Panel A: in sample (log CSV_t; Newey–West t)</i>						
predictor		(i)	(ii)	(iii)		
lagged dis		0.72 (10.6)	0.62 (4.3)	0.27 (4.7)		
lagged CSV			0.28 (5.3)	0.04 (0.7)		
lagged market RV			−0.07 (−0.9)			
lagged AIV				0.47 (8.4)		
R^2		0.30	0.35	0.41		

<i>Panel B: out of sample (OOS R^2 vs. prevailing mean; Clark–West)</i>						
subsample	n	AR+ dis	AR+AIV	AR+ dis +AIV	CW: $dis > \text{AR}$	CW: $dis > \text{AR} + \text{AIV}$
full	455	39.9%	43.3%	45.7%	7.1	4.1
ex-crisis	438	38.0%	41.4%	43.9%	6.9	3.9
expansion	415	37.1%	39.4%	42.2%	6.6	4.1
recession	40	52.8%	61.5%	61.8%	3.0	0.6

Notes. CSV_t is the cross-sectional variance of the 49 value-weighted industry returns in month t ; dis is the realized monthly volatility of η ; AIV is the average within-month daily market-model residual variance across the 49 industries. All variables in logs. Sample 1968:06–2026:04 (695 months); out-of-sample after a 240-month initial window. “Ex-crisis” drops 2008:07–2009:06 and 2020:02–2020:06; “expansion” drops NBER recession months.

IA.E Fama–MacBeth estimates and non-identification

The two-pass estimates reported in Section 8.3 of the paper regress the mean excess returns of the 25 size–book-to-market and 49 industry portfolios on first-pass betas to $[\text{Mkt} - \text{RF}, \eta]$, with Shanken (1992) standard errors. On these assets the price of market risk is estimated near zero (well below the realized premium) with a cross-sectional R^2 of about 0.03 and a large, significant intercept, and the price of η risk is indistinguishable from zero. As Lewellen et al. (2010) stress, such a cross-section does not identify risk prices, so the paper relies on the time-series alphas and the out-of-sample tests rather than on these estimates.

IA.F The three-case taxonomy

Table IA.7 summarizes the three cases of Section 1 of the paper and the evidence that separates them.

Table IA.7: Three things return predictability can be

	risk premium	mispricing	near-arbitrage
in the SDF span (priced)?	yes	no	no
gross Sharpe	bounded	positive	very large
survives net of costs?	yes	briefly	no
what erases it	nothing	arbitrage capital	friction barrier
post-publication decay	no	yes	no
economic content	marginal utility	arbitrage speed	limits to arbitrage

Notes. The three cases and the evidence that separates them. The off-frontier component is unpriced (Section 5.1 of the paper), carries a large gross Sharpe ratio but a net-of-cost Sharpe near zero (Section 6.2), and so falls in the last column, a near-arbitrage protected by frictions, not a risk premium and not a still-open mispricing.

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